Segmentation of Green Roofs in High-Resolution Remote Sensing Images With GR-Net

Zhi Wang, Xiaoyan Cao, Yao Yao, Lian Feng, and Huapeng Qin

***Abstract*—Accurately recognizing the spatial distribution of green roofs is crucial for quantitatively assessing their ecological benefits in urban areas. Deep learning (DL) has been applied to this task using remote sensing images, reducing time and labor costs. However, challenges remain due to the irregular shapes, sparse distribution, homogeneity with ground vegetation, and high annotation costs of green roofs. To address these issues, we propose an end-to-end framework for urban-scale green roof segmentation, integrating the following: 1) a high-resolution attention-based convolutional neural network (GR-Net) to extract the contours of sparsely distributed green roof patches; 2) a building guided module (BGM) to reduce mis-segmentation of ground vegetation; 3) a remote sensing prior module (RSPM) to enhance vegetation feature discrimination; and 4) data augmentation and transfer learning to improve learning efficiency and model generalization. Taking Shenzhen, Beijing, and Shanghai as case studies, we construct a diverse green roof dataset with varying sources, spectra, and spatial resolutions. On the in-domain test dataset, GR-Net achieves an $F1$ score of 0.842 and an intersection over union (IoU) of 0.744. When applied to out-of-domain test dataset from three new cities, it maintains decent performance, with an $F1$ score of 0.756 and an IoU of 0.633. We also identify the optimal configurations for each module. Overall, this work presents a practical and reliable tool for quantitative green roof assessment. The code used in our study is publicly available at https://github.com/wangzhi123321/GR-Net**

***Index Terms*—End-to-end, green roof, neural network, remote sensing, semantic segmentation.**

I. Introduction

GREEN roofs are building roofs covered with vegetation and a growing medium [1], [2]. They offer multiple environmental benefits, including mitigating urban heat island effect [3], [4], retaining stormwater [5], [6], reducing noise pollution [7], and improving air quality [8]. Due to policies promoting their adoption, the number and total area of green roofs are increasing globally [9]. However, traditional methods such as field surveys and manual interpretation of remote sensing images are labor-intensive and inefficient, resulting in incomplete urban-scale records of green roofs [10]. To overcome these limitations, an automated segmentation approach based on remote sensing images is essential.

Building an empirical model based on remote sensing prior knowledge is a fundamental approach for green roof recognition. Widely used vegetation indices, such as the normalized difference vegetation index (NDVI) [11], kernel NDVI (kNDVI) [12], and enhanced vegetation index (EVI) [13], serve as key statistical features for vegetation recognition in remote sensing images [14], [15]. To differentiate roof vegetation from ground vegetation, researchers have incorporated additional information, including roof slope and building height [16], [17], [18]. However, acquiring such data remains challenging. Furthermore, these empirical models require parameter tuning based on expert knowledge, limiting their adaptability to diverse urban environments [19].

In recent years, image semantic segmentation techniques based on deep learning (DL) have yielded significant breakthroughs across multiple domains [20], [21], including green roof segmentation. Existing approaches leverage annotated data to train neural networks, enabling direct inference of green roof distributions from remote sensing images [10], [22]. U-Net [23], a fully convolutional network with an encoder–decoder architecture, is widely employed for image segmentation tasks. ResNet [24], a deep residual network, enhances feature extraction through increased network depth. Both Simpson et al. [10] and Wu et al. [22] utilize a U-Net variant (U-Net-ResNet), replacing the encoder with ResNet. However, these methods achieve an intersection over union (IoU) of only about 0.4 in green roof segmentation, highlighting the challenges of this task.

These challenges stem from the highly irregular shapes of green roof patches, their sparse distribution, and their homogeneity with ground vegetation. Addressing the irregularity and sparse distribution of green roofs requires a segmentation model capable of capturing fine details in remote sensing images. However, the deep downsampling in U-Net-ResNet [10], [22] reduces the model's ability to preserve critical target features such as textures and contours [24]. In this approach, convolutional operations use the shared kernel to scan different receptive fields, potentially transforming background information into noise, which interferes with green roof feature

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extraction. To retain high-resolution features and accurately capture shape characteristics, integrating a high-resolution network (HRNet) [25] with an attention mechanism [26], [27] is essential. Additionally, distinguishing green roofs from homogeneous ground vegetation requires the model to differentiate roofs from floors. Existing green roof segmentation methods rely on postprocessing techniques that incorporate supplementary building information to filter out ground vegetation [10], [22], preventing end-to-end segmentation. To overcome this limitation, we propose incorporating building information as an input feature within the neural network to directly guide green roof segmentation. Moreover, most existing methods use only three RGB color channels as input [10], [22], which is insufficient for reliable green roof recognition, as green roofs do not always appear "green" [28]. Given the demonstrated success of vegetation indices like NDVI in empirical models, integrating such remote sensing prior knowledge as additional input features can help mitigate the limitations of RGB-based approaches [29].

Another major challenge in green roof semantic segmentation is the limited availability of publicly accessible annotated datasets. Due to the sparse distribution of green roofs, locating relevant samples is labor-intensive, and manual annotation is costly [30], [31], [32]. Moreover, neural networks, which contain a large number of parameters, are prone to overfitting when trained on small datasets [33], [34]. To mitigate this issue, data augmentation techniques such as geometric transformations, feature space augmentation [35], and mosaic data augmentation [36] are commonly employed to enhance sample diversity. Additionally, transfer learning enables the migration of pretrained model parameters from a source domain to a target domain, thereby reducing the need for extensive labeled samples in the target domain [37]. To improve the model's generalization on limited data, we propose integrating data augmentation with transfer learning techniques.

To address the challenges associated with green roof segmentation, this study introduces an end-to-end segmentation framework. The key contributions of this work are as follows.

1) We develop a high-resolution attention-based convolutional neural network, GR-Net, to capture fine details and address the irregular shapes and sparse distribution of green roofs.
2) We introduce a building guided module (BGM) to filter out ground vegetation for end-to-end green roof segmentation and design a remote sensing prior module (RSPM) to enhance vegetation feature extraction.
3) We construct a multisource remote sensing dataset with submeter spatial resolution annotations for building roofs and green roofs and apply data augmentation and transfer learning to mitigate the scarcity of publicly available labeled samples.
4) Experimental results demonstrate that GR-Net consistently achieves robust performance on both in-domain and out-of-domain test datasets. Additionally, extensive ablation studies, comparative analysis, input sample experiments, and threshold experiments validate the effectiveness of each module.

The remainder of this article is organized as follows. Section II describes the study area and the dataset construction process. Section III details the GR-Net architecture, small-sample techniques, experimental settings, and evaluation metrics. Section IV presents the experimental results, analyzes the limitations of GR-Net, and discusses future research directions. Finally, Section V concludes the article.

II. Study Area and Dataset Construction

A. Study Area

China has been actively promoting the "sponge city" initiative, which leverages low-impact development (LID) and green infrastructure (GI) technologies to manage urban stormwater and achieve a sustainable urban water cycle [38]. Green roofs play a crucial role in this initiative. As pilot sponge cities, Shenzhen, Beijing, and Shanghai are implementing LID strategies and accelerating green roof adoption.

This study focuses on highly urbanized areas within Shenzhen, Beijing, and Shanghai, specifically Nanshan, Futian, Luohu, Xicheng, and Jingan districts, covering a total area of approximately 362 km^2, as shown in Fig. 1. These cities are geographically distant, exhibit diverse architectural styles, and span both subtropical and north temperate climate zones, leading to varying types of green roof vegetation [39], [40], [41]. Green roofs are widely but sparsely distributed across various building types, including commercial plazas, office buildings, residential complexes, and urban villages.

B. Dataset Construction

Our dataset comprises two sources. The first consists of remote sensing images with a 0.5 m spatial resolution, acquired by the Jilin-1 Kuanfu-01 satellite between 2022 and 2023. These images are cloud-free and have undergone radiometric, sensor and geometric correction. They comprise four spectral bands: red (R), green (G), blue (B) and near-infrared (NIR), collectively denoted as RGBN. This dataset covers Nanshan, Futian, and Luohu districts in Shenzhen. The second data source is derived from ArcGIS maps captured in 2024, with a tile level of 19 and a spatial resolution of 0.6 m. These images are in RGB true-color mode and cover Xicheng district in Beijing and Jingan district in Shanghai.

As shown in Fig. 2, we use LabelMe software [42] for image cropping and annotation. The dataset includes two annotated categories: green roofs and building roofs. To ensure annotation accuracy, each label is cross-validated by two individuals. Ultimately, we construct a green roof dataset comprising 580 patches, including building roofs.

Our dataset consists of 300 training images and 280 test images, each cropped to 300×300 pixels. To assess the generalization ability of GR-Net in large-scale applications, the test dataset is divided into in-domain and out-of-domain subsets. The in-domain test dataset contains 100 images from Nanshan district (65 images) and Futian district (35 images), which are from the same regions as the training dataset. The out-of-domain test dataset includes 180 images from Luohu district (80 images), Xicheng district (50 images), and Jingan district (50 images), which differ from the training regions [43].

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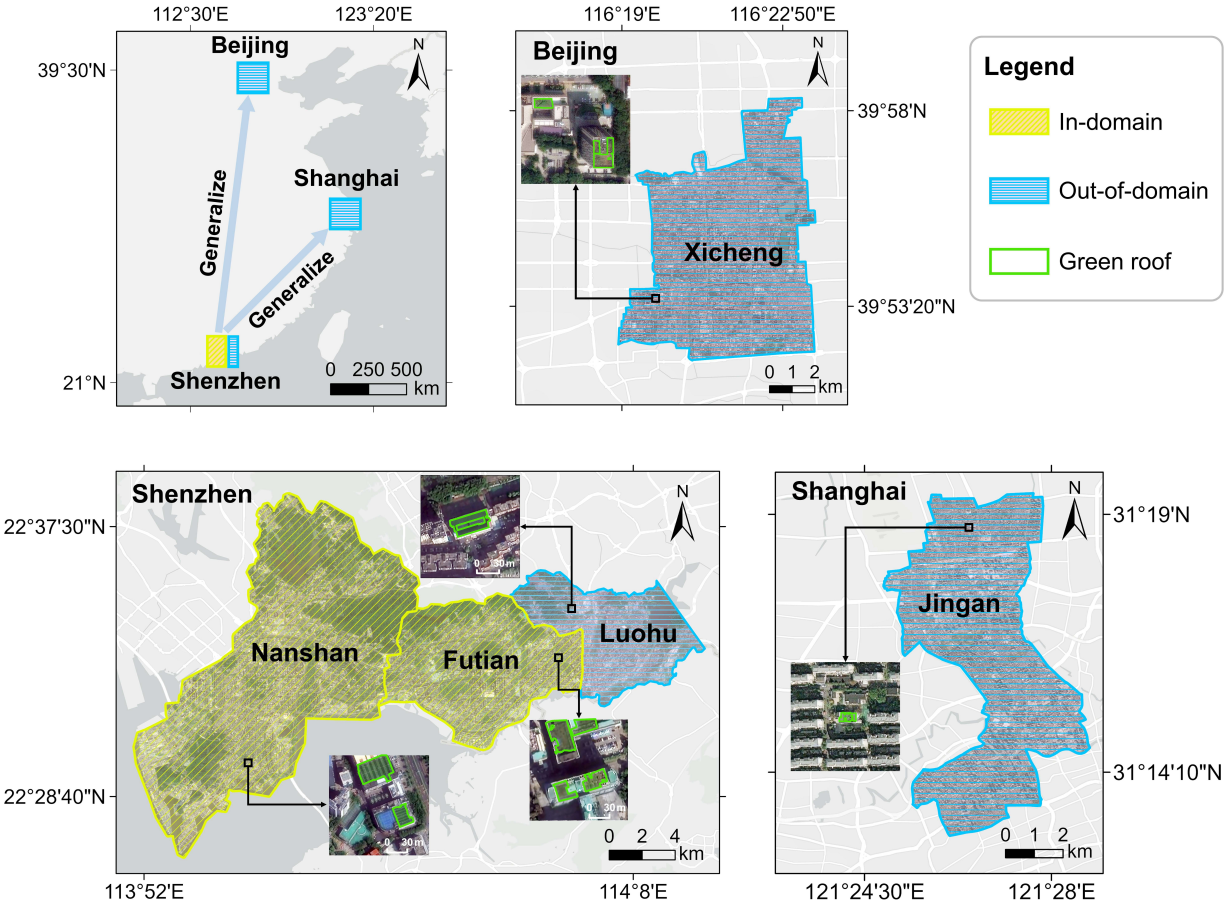


Fig. 1. Overview of the study area (the highly urbanized areas of Shenzhen, Beijing, and Shanghai in China). In-domain cities are the same regions as the training dataset, while out-of-domain cities differ from the training regions.

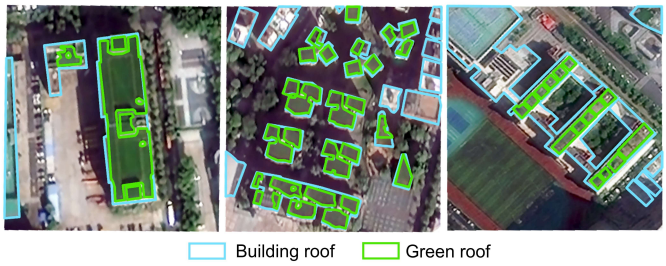


Fig. 2. Examples of labels for several remote sensing image patches. The blue box represents the building roof label. The green box represents the green roof label.

These out-of-domain cities have different architecture and vegetation styles and differ substantially from in-domain cities. In addition, the remote sensing images are taken in different seasons.

Additionally, NDVI [11], kNDVI [12], and EVI [13] are introduced to enhance the characterization of green roof vegetation features. NDVI, calculated from the NIR and R bands, ranges from -1 to 1, with higher values indicating denser and healthier vegetation. kNDVI is a nonlinear transformation of NDVI that improves robustness and sensitivity, particularly in medium to high vegetation coverage areas. EVI further incorporates the B band to correct for atmospheric aerosol scattering and includes soil adjustment factors, making it more reliable under high biomass and complex background conditions. The formulas are as follows:

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - R}{\mathrm{NIR} + R} \tag{1}$$

$$\mathrm{kNDVI} = \tanh\left(\mathrm{NDVI}^2\right) \tag{2}$$

$$\mathrm{EVI} = 2.5 \times \frac{\mathrm{NIR} - R}{\mathrm{NIR} + 6 \times R - 7.5 \times B + 1}. \tag{3}$$

To accelerate the convergence of the neural network, we normalized the RGBN, NDVI, kNDVI, and EVI values as follows:

$$\mathrm{RGBN}_{\mathrm{norm}} = \frac{\mathrm{RGBN}}{255.0} \tag{4}$$

$$\mathrm{NDVI}_{\mathrm{norm}} = \frac{\mathrm{NDVI} + 1}{2.0} \tag{5}$$

$$\mathrm{kNDVI}_{\mathrm{norm}} = \frac{\mathrm{kNDVI} - \min\left(\mathrm{kNDVI}\right)}{\max\left(\mathrm{kNDVI}\right) - \min\left(\mathrm{kNDVI}\right)} \tag{6}$$

$$\mathrm{EVI}_{\mathrm{norm}} = \frac{\mathrm{EVI} - \min\left(\mathrm{EVI}\right)}{\max\left(\mathrm{EVI}\right) - \min\left(\mathrm{EVI}\right)}. \tag{7}$$

III. Methods

A. GR-Net

As shown in Fig. 3(a), we propose GR-Net, an end-to-end neural network-based method for green roof segmentation.

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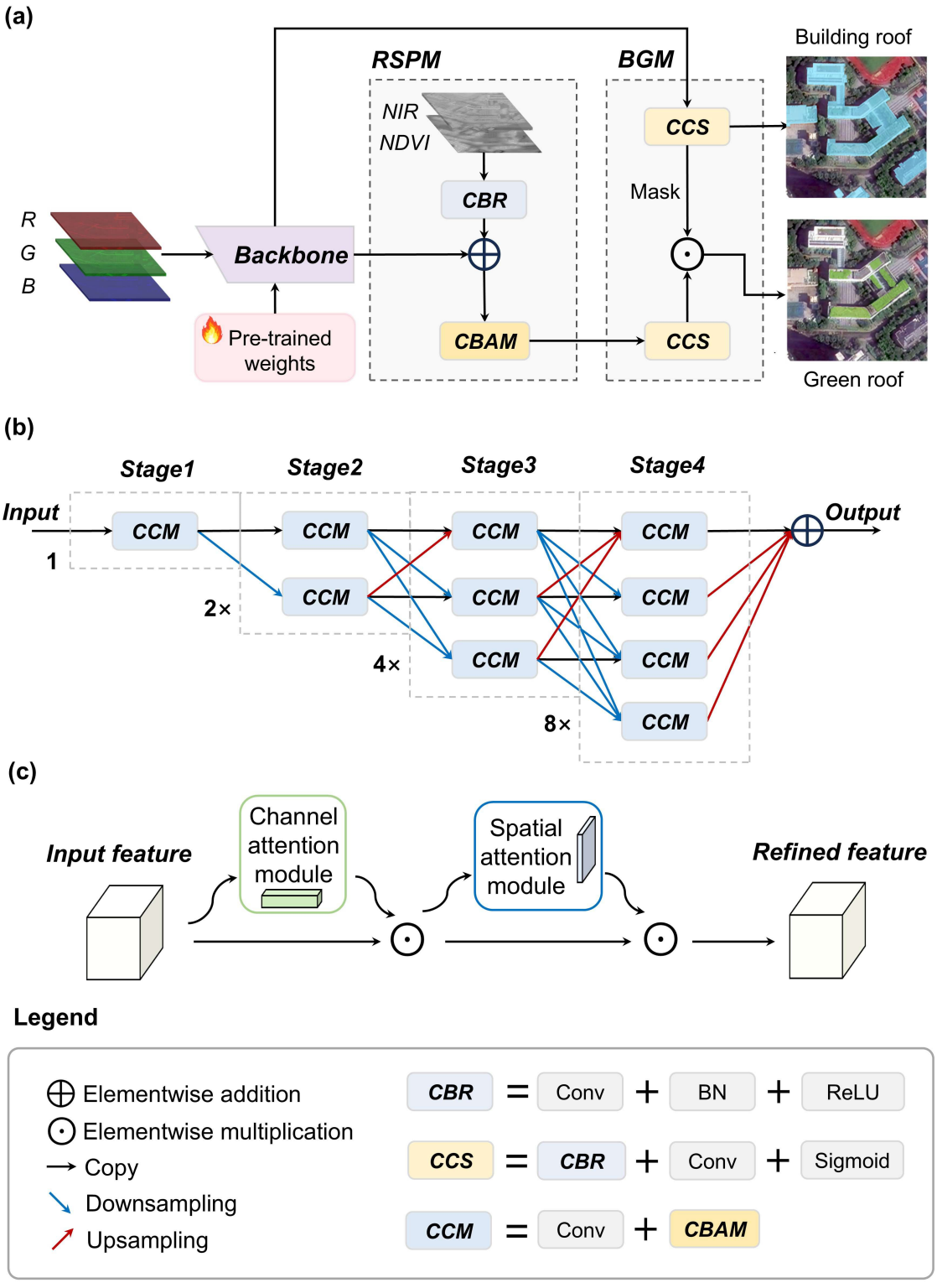


Fig. 3. Architecture of GR-Net. (a) Overview of GR-Net, consisting of a backbone, RSPM, and BGM. The backbone receives RGB images as input and pretrained weights as parameters for initialization. The backbone is followed by two neural network modules, RSPM and BGM, which are used for predicting green roofs and building roofs. While RSPM receives remote sensing prior information (NIR and NDVI) to enhance feature extraction for vegetation, BGM receives outputs from backbone and RSPM for filtering ground vegetation. (b) Structure of backbone, which is an improved HRNet [25] integrating CBAM [44]. The backbone consists of four stages, comprising parallel high-resolution to low-resolution subnetworks with repeated information exchange across multiresolution subnetworks. (c) Structure of CBAM. The module has two sequential submodules: channel attention module (CAM) and spatial attention module (SAM).

Sections III-A1–III-A4 provide a detailed description of each module.

1) Backbone of GR-Net: We propose a novel backbone that integrates the convolutional block attention module (CBAM)

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[44] with HRNet [25]. As illustrated in Fig. 3(b), the backbone comprises a series of Conv-CBAMs (CCMs) that combine convolutional layers (Convs) with CBAMs, enabling multiresolution convolution streams to operate in parallel. The CBAM structure, depicted in Fig. 3(c), integrates two key attention mechanisms: channel attention mechanism and spatial attention mechanism. CAM enhances feature selection by prioritizing relevant channels, whereas SAM emphasizes critical spatial regions. HRNet preserves high-resolution feature maps throughout the network and captures multiscale contextual information by fusing features across different resolutions.

2) Remote Sensing Prior Module: Inspired by the traditional empirical model, we use remote sensing prior information (i.e. NIR and NDVI) as input variables of the RSPM to enhance the feature extraction of vegetation in the remote sensing images, as depicted in Fig. 3(a) in the revised manuscript. These inputs are processed through a Conv-BN-ReLU (CBR) block, which consists of a 3×3 convolutional layer (Conv), a batch normalization (BN) [45], and a rectified linear unit (ReLU) [46] activation function. The RSPM's output feature map is fused with the output features of the backbone, which only employs an RGB image as an input, to enhance the recognition of vegetation. Additionally, CBAM is incorporated to further focus on the vegetation features.

3) Building Guided Module: Since there is a strong prior knowledge that green roofs must be on the buildings roofs, a learnable BGM is designed to reduce false positives (FPs) from nonroof vegetation. As illustrated in Fig. 3(a), the feature maps obtained from the backbone and RSPM are processed through CBR-Conv-Sigmoid (CCS) modules to segment building roofs and green roofs, respectively. These CCS modules process the feature maps to generate the final segmentation outputs. To distinguish green roofs from ground vegetation, the segmentation output for building roof, denoted as x_b, is used as a mask to filter out nonroof areas from the green roof segmentation result x_g, as expressed in the following equation:

$$x_g = x_b \times \text{mask} \tag{8}$$

$$\text{mask} = \begin{cases} 1, & \text{if } x_b \geq \text{threshold} \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

where threshold represents the threshold of the mask (ranging from 0 to 1, set to 0.5 in this study).

4) Loss Function: The loss function for training GR-Net comprises two components: building roof loss and green roof loss. Both are formulated as a combination of focal loss [47] and Dice loss [48]. Focal loss is designed to mitigate class imbalance by assigning greater weight to hard-to-classify samples, thereby improving the model's ability to distinguish minority class instances. It is defined as follows:

$$\mathcal{L}_{\text{focal}}\left(F_i, \hat{F}_i\right) = \frac{1}{N}\sum_{i=1}^{N}\left[-\alpha\left(1-\hat{F}_i\right)^{\gamma} F_i \log\left(\hat{F}_i\right) - (1-\alpha)\,\hat{F}_i^{\gamma}\left(1-F_i\right)\log\left(1-\hat{F}_i\right)\right] \tag{10}$$

where $\mathcal{L}_{\text{focal}}(F_i, \hat{F}_i)$ denotes the focal loss, F_i denotes the ith pixel value of the ground truth, and $\hat{F}_i$ denotes the ith pixel value of the predicted value. $\alpha \in [0, 1]$ is a parameter used to balance the importance of positive and negative samples, which is taken as 0.25 in this study; $\gamma > 0$ denotes the moderating factor, which is taken as 2 in this article; N denotes the total number of pixels.

Dice loss is widely used in segmentation tasks as it enhances accuracy by optimizing the overlap between predicted and ground truth regions. It is defined as follows:

$$\mathcal{L}_{\text{dice}}\left(F_i, \hat{F}_i\right) = 1 - \frac{2\sum_{i=1}^{N} F_i \hat{F}_i}{\sum_{i=1}^{N} F_i + \sum_{i=1}^{N} \hat{F}_i} \tag{11}$$

where $\mathcal{L}_{\text{dice}}(F_i, \hat{F}_i)$ denotes dice loss.

The loss function for building roofs is expressed as follows:

$$\mathcal{L}_b = \lambda_b \cdot \mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{dice}} \tag{12}$$

where λ_b denotes a weighting factor of building roofs.

Similarly, the loss function for green roofs is expressed as follows:

$$\mathcal{L}_g = \lambda_g \cdot \mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{dice}} \tag{13}$$

where λ_g denotes a weighting factor of green roofs.

The loss of GR-Net is expressed as follows:

$$\mathcal{L} = \beta\mathcal{L}_b + \mathcal{L}_g \tag{14}$$

where β is used to balance the loss of building roofs and green roofs.

B. Small Sample Techniques

1) Data Augmentation: By leveraging data augmentation, we enhance the model's generalization performance despite the limited number of supervised samples. Specifically, we employ the mosaic data augmentation technique from YOLOv4 [36]. As illustrated in Fig. 4, the process begins with randomly selecting four images from the training dataset. These images undergo a series of transformations, including flipping (horizontal or vertical), rotation (0 to 360°), resizing (scale of 0.5 to 1.0), random cropping (size of 150×150), and modifications to brightness ($\times$0.9 to 1.1), contrast ($\times$0.9 to 1.1), saturation ($\times$0.9 to 1.1), and hue ($-10°$ to $+10°$). The processed images are then merged to create a new composite image. Through this approach, the training dataset is expanded from 300 samples to 10 000, significantly improving the model's robustness and generalization capability.

2) Transfer Learning: We utilize transfer learning to enhance the efficiency of GR-Net in learning from limited samples. Specifically, HRNet in the backbone has been widely used for modeling various tasks, including sufficient training on two large-scale datasets, PASCAL VOC and SBD. These datasets encompass a broad range of object classes and complex scenes, including irregular shapes, overlapping objects, and intricate backgrounds. These challenges are very similar to those in green roof segmentation. By leveraging the pretrained HRNet parameters, we initialize the corresponding parameters in GR-Net, enabling the model to inherit useful feature representations. The remaining parameters are initialized using PyTorch's default methods, such as Kaiming initialization [49], which is well suited for networks utilizing ReLU activation functions.

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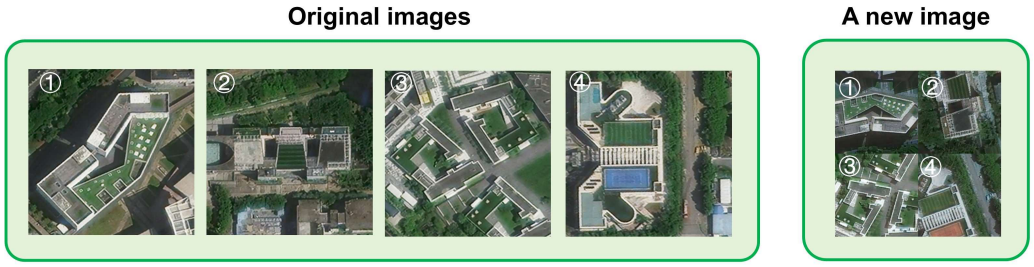


Image	Flip	Rotate	Resize	Crop	Brightness	Contrast	Hue	Saturation
①	Horizontal	232°	58%	25%	91%	103%	0°	95%
②	Non	283°	81%	25%	99%	98%	-6°	93%
③	Horizontal	199°	68%	25%	109%	108%	-5°	91%
④	Vertical	199°	72%	25%	94%	102%	-5°	90%

Fig. 4. Example of data augmentation. A new image is generated by performing a series of operations such as flipping, rotation, resizing, random cropping, and modifications to brightness, contrast, hue, and saturation on each of the four original images.

C. Experimental Settings

To evaluate the effectiveness of GR-Net in green roof segmentation, we conduct the following experiments.

1) *Generalization Experiments*: We assess the model's ability to segment green roofs across different urban scales by testing it on both in-domain and out-of-domain datasets.
2) *Ablation Studies*: We analyze the contribution of the RSPM, CCS, BGM, and CBAM components by adding them gradually or individually to the baseline model and observing their impact on the segmentation performance.
3) *Method Comparison Studies*: We compare various state-of-the-art methods on the segmentation performance, including HRNet [25], U-Net [23], U-Net-ResNet [24], U-Net-VGG [50], DeepLab-MobileNet [51], PSPNet-MobileNet [52], PSPNet-ResNet [52], SegFormer [53], TransUNet [54], TransFuse [55], and Mobile-Former [56].
4) *Attention Mechanism Comparison Studies*: We examine the impact of different attention mechanisms, including CBAM [44], CANet [57], CCNet [58], SENet [59], SimAM [60], ECANet [61], NAM [62], and DANet [63], on segmentation performance.
5) *Input Sample Studies*: These experiments explore various factors affecting segmentation accuracy: 1) the influence of remote sensing prior knowledge by comparing vegetation indices (NDVI [11], kNDVI [12], and EVI [13]); 2) the effectiveness of data augmentation by varying the degree of mosaic data augmentation; the impact of training sample size by expanding randomly selected subsets of the original training dataset to 10 000 samples; 3) the role of transfer learning by comparing models with and without pretrained weights under different training sample sizes; and 4) the effect of spatial resolution variations on segmentation performance.
6) *Threshold Studies*: We adjust the threshold of BGM from 0.1 to 0.9 in steps of 0.2 and observe their impact on model performance.

For a fair comparison, all models in the above experiments are run in the same computing environment. The experiments are conducted on a system with a 48-core AMD 7543 CPU and 8 NVIDIA GeForce RTX 4090 GPUs running Linux. All neural network frameworks utilize PyTorch. During training, we employ the Adam optimizer with a learning rate that ranges from 1×10^{-3} to 5×10^{-6}, adjusted using the cosine annealing strategy. The number of training epochs is set to 50, with a batch size of 70. To ensure reproducibility, we fix the random seed at 42.

D. Evaluation Metrics

We use four metrics, $F1$ score, IoU, Boundary $F1$ score [64], and Boundary IoU [65], to evaluate the accuracy of GR-Net.

The $F1$ score measures the accuracy of a binary classification model, ranging from 0 to 1, where higher values indicate better predictions. It accounts for both Precision and Recall, providing a balanced evaluation of the model's performance. The equation for the $F1$ score is as follows:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

where Precision and Recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

where true positive (TP) denotes the number of perfectly predicted green roof pixels; FP denotes the number of wrongly predicted green roof pixels as positives that are not green roofs; false negative (FN) denotes the number of wrongly predicted nongreen roof pixels that are green roofs.

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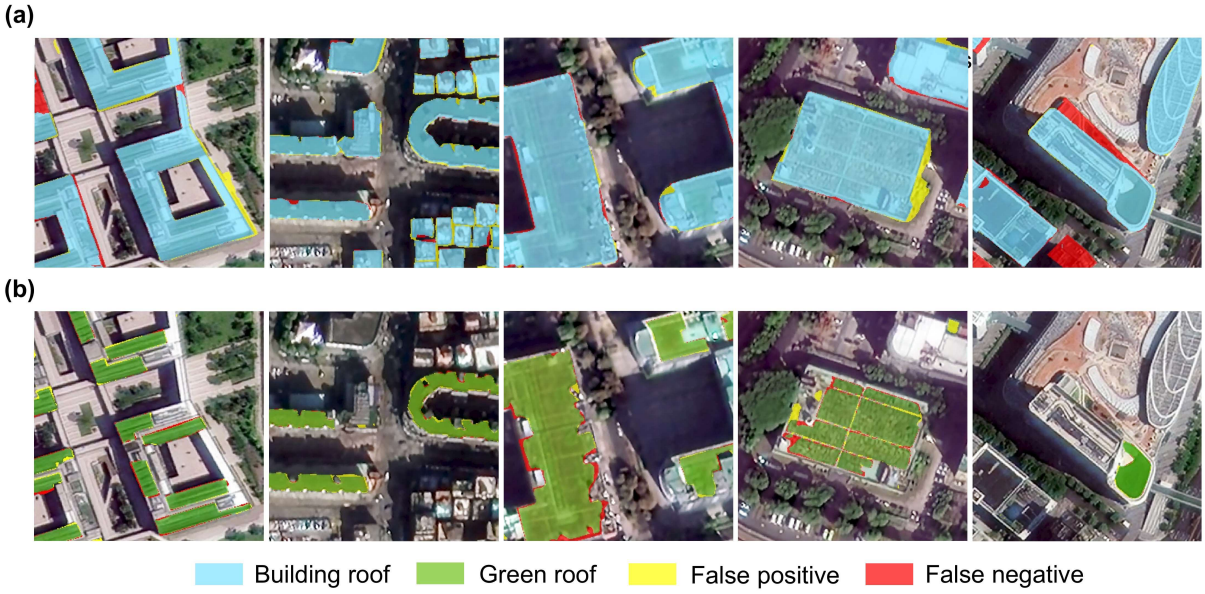


Fig. 5. Prediction results of GR-Net on in-domain test dataset. (a) Building roof and (b) green roof prediction results.

IoU is used to calculate the degree of overlap between the predicted value and the ground truth, with values ranging from 0 to 1. The larger the value, the more accurate the prediction. IoU is calculated as follows:

$$\text{IoU} \models \frac{|\hat{F} \cap F|}{|\hat{F} \cup F|} \tag{18}$$

where $\hat{F}$ and F denote the predicted value and ground truth, respectively.

Boundary $F1$ score [64] is an improvement of $F1$ score to evaluate the model's ability to capture boundaries. The boundary is first extracted from ground truth mask and predicted mask and the distance between boundary points is calculated. Then Boundary Precision (BP) is calculated to measure how many predicted boundary pixels match the true boundary pixels. Boundary Recall (BR) is calculated to measure how many true boundary pixels are correctly predicted as boundaries. The equation for the BF score is as follows:

$$\text{Boundary } F1 = \frac{2 \cdot \text{BP} \cdot \text{BR}}{\text{BP} + \text{BR}} \tag{19}$$

where BP and BR are defined as follows:

$$\text{BP} = \frac{\left|\left\{p \in \partial P \mid \min_{g \in \partial G} d(p,g) \le \theta\right\}\right|}{|\partial P|} \tag{20}$$

$$\text{BR} = \frac{\left|\left\{g \in \partial G \mid \min_{p \in \partial P} d(g,p) \le \theta\right\}\right|}{|\partial G|} \tag{21}$$

where ∂G is the set of boundary points extracted from the ground truth mask; ∂P is the set of boundary points extracted from the predicted mask; $d(p,g)$ is the Euclidean distance between boundary points p and g; and θ is the distance tolerance, typically defined as $\theta = \alpha \cdot (H^2 + W^2)^{1/2}$, where α is a predefined dilation ratio (set to 0.0075 in this study), H and W represent the height and width of the mask, respectively.

Boundary IoU [65] is a measure of the proportion of overlap between the boundary of ground truth mask and predicted mask, which extends IoU to account for boundary dilation. The equation is as follows:

$$\text{Boundary IoU} = \frac{|(G_d \cap G) \cap (P_d \cap P)|}{|(G_d \cap G) \cup (P_d \cap P)|} \tag{22}$$

where G is the binary boundary mask extracted from the ground truth mask; P is the binary boundary mask extracted from the predicted mask; G_d is the dilation of G using a tolerance radius $r = \alpha \cdot (H^2 + W^2)^{1/2}$, where α is a predefined dilation ratio (set to 0.02 in this study), H and W represent the height and width of the mask, respectively; and P_d is the dilation of P using the same radius.

IV. Results and Discussion

A. Generalizability of GR-Net

The in-domain test dataset includes Nanshan and Futian districts, with input data consisting of five channels: R, G, B, NIR, and NDVI, at a resolution of 0.5 m. As shown in Table I and Fig. 5, GR-Net recognizes nearly all building roofs, which effectively assists in the accurate segmentation of green roofs.

Next, we apply GR-Net to the out-of-domain test dataset. The input data for Luohu district consists of five channels: R, G, B, NIR, and NDVI at 0.5 m resolution, while Xicheng and Jingan districts use three channels: R, G, and B at 0.6 m resolution. As shown in Table I, the performance of GR-Net in the out-of-domain cities is still acceptable although it is slightly degraded compared to the in-domain cities. As shown in Fig. 6, the visualization results indicate that GR-Net accurately segments both the number and area of green roofs. These findings demonstrate the model's strong generalization ability across different green roof styles, suggesting its effectiveness for applications in other cities.

Fig. 7 illustrates the distribution of green roofs predicted by GR-Net in the study area. We then calculate the predicted area

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TABLE I

PREDICTION RESULTS OF GR-NET ON THE IN-DOMAIN AND OUT-OF-DOMAIN TEST DATASETS

Domain type	Roof type	*F*1	*IoU*	*Boundary F*1	*Boundary IoU*
In-domain (Nanshan, Futian)	Building roof	0.900	0.822	0.762	0.726
	Green roof	0.842	0.744	0.816	0.731
Out-of-domain (Luohu, Xicheng, Jingan)	Building roof	0.816	0.700	0.648	0.615
	Green roof	0.756	0.633	0.726	0.629

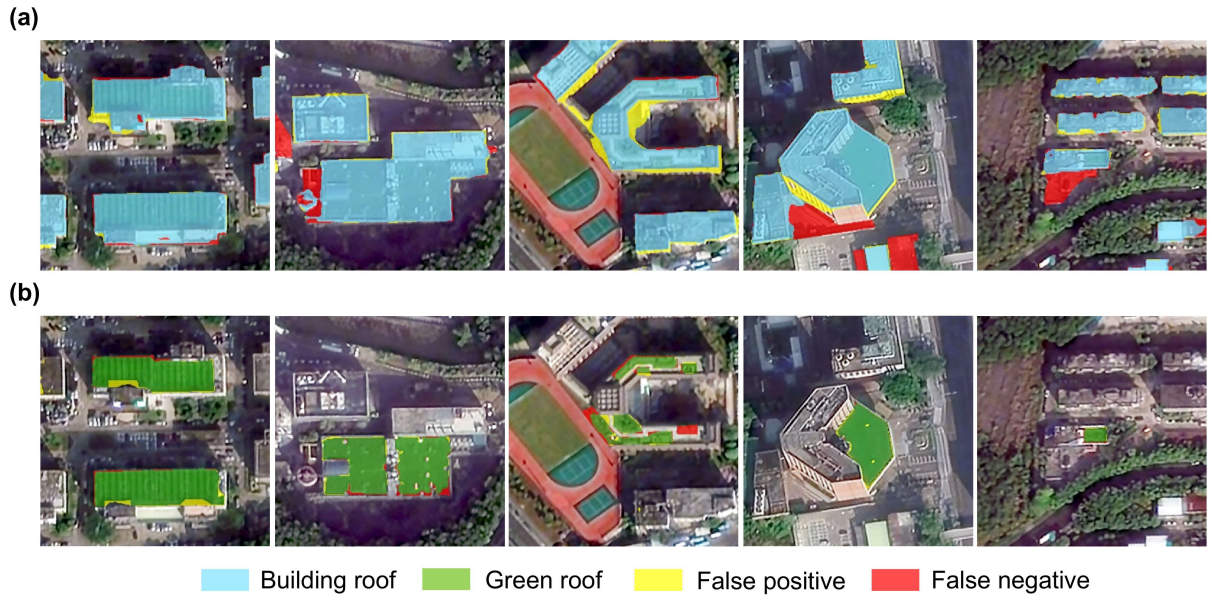


Fig. 6. Prediction results of GR-Net on out-of-domain test dataset. (a) Building roof and (b) green roof prediction results.

TABLE II

STEPWISE ABLATION STUDY OF GR-NET. THE GREEN VALUES IN PARENTHESES INDICATE THE IMPROVEMENT COMPARED TO THE BASELINE

Method	*F*1	*IoU*	*Boundary F*1	*Boundary IoU*
Baseline (= HRNet)	0.768	0.664	0.743	0.650
Baseline + RSPM	0.801 (↑4.3%)	0.697 (↑5.0%)	0.776 (↑4.4%)	0.683 (↑5.1%)
Baseline + RSPM + CCS	0.813 (↑5.9%)	0.709 (↑6.8%)	0.781 (↑5.1%)	0.695 (↑6.9%)
Baseline + RSPM + CCS + BGM	0.829 (↑7.9%)	0.725 (↑9.2%)	0.799 (↑7.5%)	0.712 (↑9.5%)
Baseline + RSPM + CCS + BGM + CBAM	0.842 (↑9.6%)	0.744 (↑12.0%)	0.816 (↑9.8%)	0.731 (↑12.5%)

TABLE III

SINGLE-STEP ABLATION STUDY OF GR-NET ON THE IN-DOMAIN TEST DATASET. THE GREEN VALUES IN PARENTHESES INDICATE THE IMPROVEMENT COMPARED TO THE BASELINE

Method	*F*1	*IoU*	*Boundary F*1	*Boundary IoU*
Baseline (= HRNet)	0.768	0.664	0.743	0.650
Baseline + BGM	0.815 (↑6.1%)	0.707 (↑6.5%)	0.787 (↑5.9%)	0.695 (↑6.9%)
Baseline + RSPM	0.801 (↑4.3%)	0.697 (↑5.0%)	0.776 (↑4.4%)	0.683 (↑5.1%)
Baseline + CBAM	0.795 (↑3.5%)	0.691 (↑4.1%)	0.767 (↑3.2%)	0.679 (↑4.5%)
Baseline + CCS	0.785 (↑2.2%)	0.675 (↑1.7%)	0.760 (↑2.3%)	0.664 (↑2.2%)

and proportion of building roofs and green roofs. The results indicate that building roofs cover 54.66 km^2, accounting for 15.10% of the total study area, while green roofs cover 2.95 km^2, representing 0.81% of the total area and 5.40% of the building roof area. At the subdistrict level, the green roof areas in Nanshan, Futian, Luohu, Xicheng, and Jingan districts are 1.01, 0.70, 0.32, 0.62, and 0.30 km^2, respectively. These findings provide valuable scientific insights for the planning and management of sponge city construction.

B. Ablation Study

We validate the effectiveness of each module through stepwise and single-step ablation studies, as shown in Tables II and III and Fig. 8.

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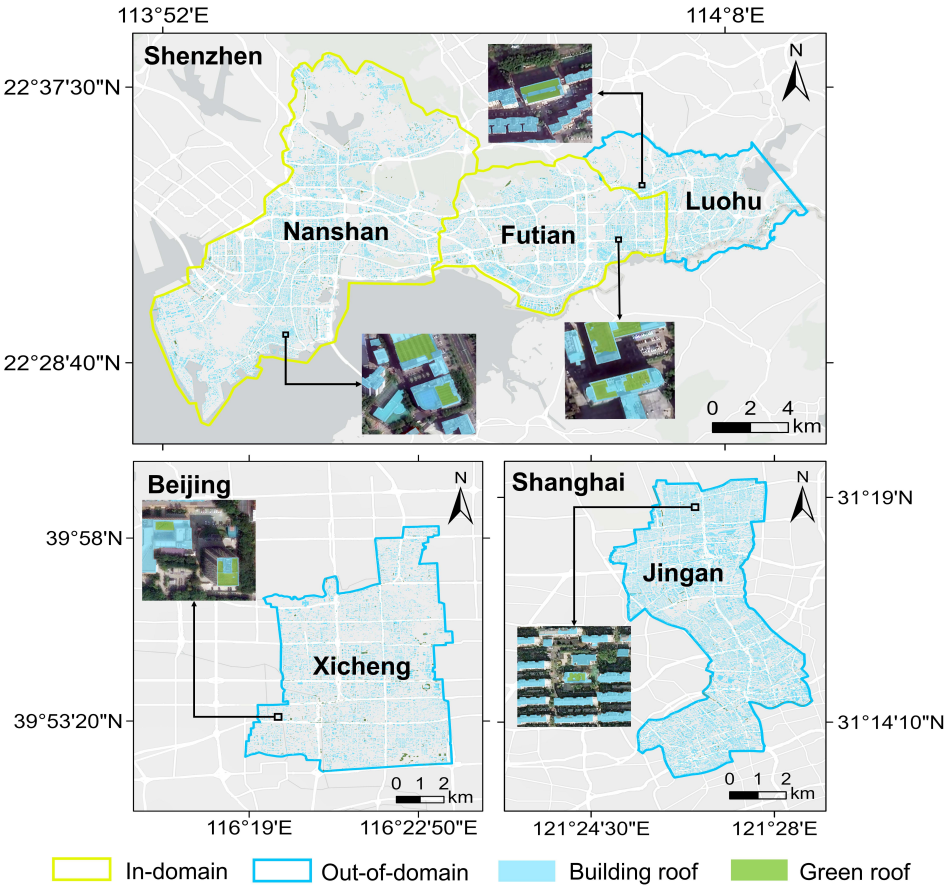


Fig. 7. Distribution of green roofs predicted by GR-Net in study area. In-domain cities include Nanshan and Futian districts in Shenzhen, while out-of-domain cities include Luohu district in Shenzhen, Xicheng district in Beijing, and Jingan district in Shanghai.

1) *RSPM*: As shown in Tables II and III, incorporating RSPM into the baseline significantly improves model performance. $F1$ score, IoU, Boundary $F1$, and Boundary IoU improve by 4.3%, 5.0%, 4.4%, and 5.1% respectively. The visualization in Fig. 8(b) demonstrates the effectiveness of remote sensing prior knowledge in capturing vegetation features.
2) *CCS*: In the stepwise ablation study, the CCS module is incorporated into both the building roof and green roof prediction branches. As shown in Table II, compared to the baseline, $F1$ score, IoU, Boundary $F1$, and Boundary IoU improve by 5.9%, 6.8%, 5.1%, and 6.9%, respectively. Moreover, in the single-step ablation study, as shown in Table III, incorporating CCS into the baseline alone also improves model performance, with the $F1$ score, IoU, Boundary $F1$, and Boundary IoU improving by 2.2%, 1.7%, 2.3%, and 2.2%, respectively. In terms of visualization, Fig. 8(c) shows that CCS effectively integrates multiscale features, improving the model's ability to capture both global semantics and local details.
3) *BGM*: In the stepwise ablation study, as shown in Table II, compared to the baseline, incorporating BGM improves the $F1$ score, IoU, Boundary $F1$, and Boundary IoU by 7.9%, 9.2%, 7.5%, and 9.5%, respectively. In the single-step ablation study, as shown in Table III, the addition of BGM to the baseline results in the greatest improvement in model performance, with an increase of 6.1%, 6.5%, 5.9%, and 6.9% in the $F1$ score, IoU, Boundary $F1$ score, and Boundary IoU, respectively. As shown in Fig. 8(d), BGM effectively filters out nonroof areas, such as ground vegetation, enabling end-to-end accurate semantic segmentation of green roofs.
4) *CBAM*: On the one hand, in the stepwise ablation study, incorporating CBAM into the model completes the GR-Net architecture. As shown in Table II, compared to the baseline, GR-Net achieves an improvement of 9.6%, 12.0%, 9.8%, and 12.5% in the $F1$ score, IoU, Boundary $F1$ score, and Boundary IoU, respectively. On the other hand, in the single-step ablation study, as shown in Table III, incorporating CBAM alone improves the $F1$ score, IoU, Boundary $F1$, and Boundary IoU by 3.5%, 4.1%, 3.2%, and 4.5%, respectively. Furthermore, Fig. 8(e) illustrates a significant enhancement in the model's ability to segment green roofs in noisy backgrounds, especially in the boundary. Additionally, as shown in Table IV, we test various combinations of

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TABLE IV

ABLATION STUDY OF CBAM ON THE IN-DOMAIN TEST DATASET. THE GREEN VALUES IN PARENTHESES INDICATE THE IMPROVEMENT COMPARED TO THE BASELINE. THE RED VALUES IN PARENTHESES INDICATE THE REDUCTION COMPARED TO THE BASELINE

Method	*F*1	*IoU*	*Boundary F*1	*Boundary IoU*
Baseline	0.768	0.664	0.743	0.650
Baseline + CAM + SAM in sequence (= CBAM)	0.795 (↑3.5%)	0.691 (↑4.1%)	0.767 (↑3.2%)	0.679 (↑4.5%)
Baseline + SAM + CAM in sequence	0.774 (↑0.8%)	0.663 (↓0.2%)	0.745 (↑0.3%)	0.651 (↑0.2%)
Baseline + CAM + SAM in parallel	0.774 (↑0.8%)	0.669 (↑0.8%)	0.747 (↑0.5%)	0.658 (↑1.2%)
Baseline + CAM (= CBAM w/o SAM)	0.783 (↑2.0%)	0.676 (↑1.8%)	0.752 (↑1.2%)	0.664 (↑2.2%)
Baseline + SAM (= CBAM w/o CAM)	0.784 (↑2.1%)	0.681 (↑2.6%)	0.750 (↑0.9%)	0.670 (↑3.1%)

Baseline: GR-Net w/o CBAM

TABLE V

COMPARISON WITH STATE-OF-THE-ART SEMANTIC SEGMENTATION METHODS ON THE IN-DOMAIN TEST DATASET. THE BEST VALUE IS SHOWN IN RED, AND THE SECOND-BEST VALUE IS SHOWN IN BLUE

Method	Journal (Year)	*F*1	*IoU*	*Boundary F*1	*Boundary IoU*
U-Net-VGG [50]	ICLR (2015)	0.484	0.338	0.333	0.323
U-Net-ResNet [10], [22], [24]	CVPR (2016)	0.525	0.376	0.349	0.361
Mobile-Former [56]	CVPR (2022)	0.654	0.513	0.407	0.464
PSPNet-MobileNet [52]	CVPR (2017)	0.665	0.524	0.493	0.496
SegFormer [53]	NeurIPS (2021)	0.670	0.537	0.598	0.525
TransUNet [54]	MedIA (2024)	0.724	0.593	0.641	0.578
DeepLab-MobileNet [51]	CVPR (2017)	0.731	0.600	0.654	0.583
PSPNet-ResNet [52]	CVPR (2017)	0.759	0.633	0.649	0.613
TransFuse [55]	MICCAI (2021)	0.765	0.648	0.698	0.631
U-Net [23]	MICCAI (2015)	0.769	0.655	0.717	0.640
HRNet [25]	TPAMI (2020)	**0.819**	**0.713**	**0.789**	**0.698**
GR-Net (Ours)	–	**0.842**	**0.744**	**0.816**	**0.731**

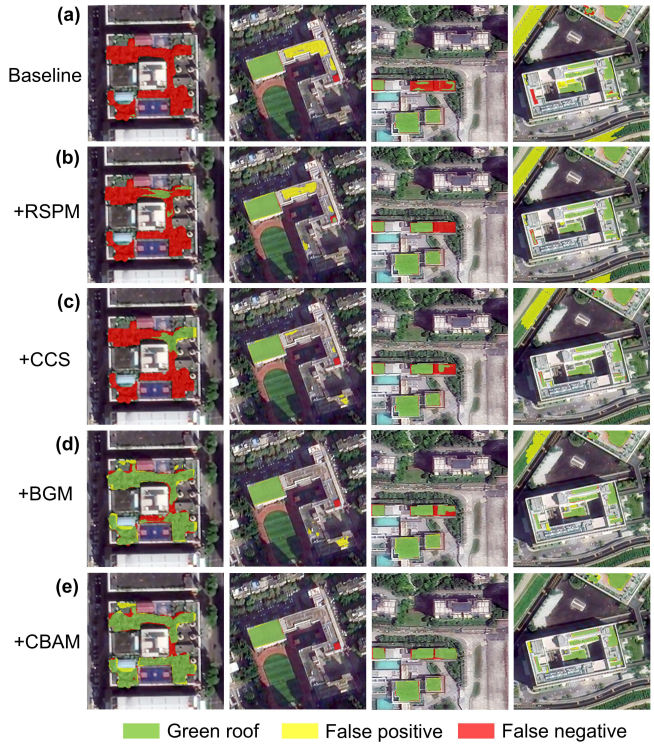


Fig. 8. Comparison of stepwise ablation study results. (a) Prediction results of baseline. (b)–(e) Prediction results after adding RSPM, CCS, BGM, and CBAM module by module. Baseline: GR-Net without RSPM, CCS, BGM, and CBAM.

CAM and SAM. The results confirm that the sequence concatenation of CAM and SAM (i.e., CBAM) achieves the best performance across all metrics on the in-domain test dataset.

TABLE VI

COMPARISON OF DIFFERENT ATTENTION MECHANISMS ON THE IN-DOMAIN TEST DATASET. THE BEST VALUE IS SHOWN IN RED, AND THE SECOND-BEST VALUE IS SHOWN IN BLUE

Attention mechanism	*F*1	*IoU*	*Boundary F*1	*Boundary IoU*
CANet [57]	0.804	0.696	0.767	0.683
CCNet [58]	0.816	0.706	0.759	0.691
SENet [59]	0.823	0.721	0.792	0.705
SimAM [60]	0.826	0.719	0.784	0.708
ECANet [61]	0.830	0.726	**0.798**	0.714
NAM [62]	0.831	0.724	0.791	0.711
DANet [63]	**0.837**	**0.732**	0.793	**0.718**
CBAM [44]	**0.842**	**0.744**	**0.816**	**0.731**

C. Comparison With State-of-the-Art Methods

We compare eleven state-of-the-art semantic segmentation methods. As shown in Table V, GR-Net outperforms the other methods such as U-Net, SegFormer, and TransFuse across all metrics on the in-domain test dataset. This superior performance is attributed to the fact that GR-Net not only loads pretrained weights, but also incorporates HRNet and CBAM, which allows it to capture fine details and address the irregular shapes and sparse distribution of green roofs.

D. Comparison of Attention Mechanisms

We evaluate eight common attention mechanisms. As shown in Table VI, CBAM outperforms the other attention mechanisms such as CANet, SimAM, and DANet on the in-domain

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TABLE VII

COMPARISON OF DIFFERENT BOUNDARY REGULARIZATION METHODS ON THE IN-DOMAIN TEST DATASET. THE BEST VALUE IS SHOWN IN RED, AND THE SECOND-BEST VALUE IS SHOWN IN BLUE

Method	*F1*	*IoU*	*Boundary F1*	*Boundary IoU*
Baseline	0.829	0.725	0.799	0.712
Baseline + CBAM [44]	**0.842**	**0.744**	**0.816**	**0.731**
Baseline + Boundary Loss [66]	0.833	0.728	0.801	0.714
Baseline + BEM [67]	0.747	0.618	0.640	0.598
Baseline + CBAM [44] + Boundary Loss [66]	**0.837**	**0.733**	**0.810**	**0.722**
Baseline + CBAM [44] + BEM [67]	0.781	0.663	0.701	0.647

Baseline: GR-Net w/o CBAM
Boundary Loss: A loss function for supervising boundary

test dataset. This is because CBAM effectively enhances feature recognition by simultaneously capturing channel and spatial information. Therefore, CBAM is used as the attention mechanism in all our experiments.

E. Comparison of Boundary Regularization Methods

We compare two state-of-the-art boundary regularization methods, Boundary Loss [66] and Boundary Enhancement Module (BEM) [67], with our attention mechanism CBAM. As shown in Table VII, compared to the baseline, BEM leads to a decrease in model performance, whereas both Boundary Loss and CBAM significantly improve performance, with CBAM achieving the highest scores across all metrics on the in-domain test dataset.

We further investigate the compatibility between CBAM and Boundary Loss. The results show that no significant improvement is observed when the Boundary Loss is added on top of CBAM. This is because CBAM already enables accurate boundary segmentation, leaving little room for further gains from explicit boundary refinement. Moreover, introducing an additional boundary-specific network diverts part of the backbone's capacity away from the primary task of green roof segmentation.

These findings not only confirm the effectiveness of CBAM in handling boundary information but also highlight the innovation of our GR-Net, which integrates HRNet with CBAM for superior overall performance without the need for additional boundary refinement modules.

F. Effectiveness of Input Samples

This section examines the impact of input samples on model accuracy, focusing on remote sensing prior knowledge, data augmentation, transfer learning, and image spatial resolution.

1) Remote Sensing Prior Knowledge: To evaluate the influence of remote sensing prior knowledge on green roof semantic segmentation, we compare three vegetation indices—NDVI [11], kNDVI [12], and EVI [13]—against models trained solely on RGB or RGBN images (i.e., without remote sensing prior knowledge).

As shown in Table VIII, introducing the NIR band in RSPM leads to only a limited improvement in model performance. Incorporating EVI as prior knowledge cannot improve model performance, which may be due to the fact that EVI overfits the parameters for the previous study case [13] and is not applicable to all vegetation types. In contrast, both NDVI and kNDVI result in significant improvements in green roof segmentation, especially NDVI improves from 3% to 5%. Therefore, NDVI is adopted as the remote sensing prior feature in subsequent experiments.

TABLE VIII

COMPARISON OF DIFFERENT REMOTE SENSING PRIOR KNOWLEDGE ON THE IN-DOMAIN TEST DATASET. THE BEST VALUE IS SHOWN IN RED, AND THE SECOND-BEST VALUE IS SHOWN IN BLUE

Input of Backbone	Input of RSPM	*F1*	*IoU*	*Boundary F1*	*Boundary IoU*
RGB	–	0.809	0.705	0.775	0.692
	NIR	0.817	0.712	0.788	0.700
	NIR + EVI [13]	0.817	0.714	0.787	0.700
	NIR + kNDVI [12]	**0.834**	**0.728**	**0.797**	**0.713**
	NIR + NDVI [11]	**0.842**	**0.744**	**0.816**	**0.731**

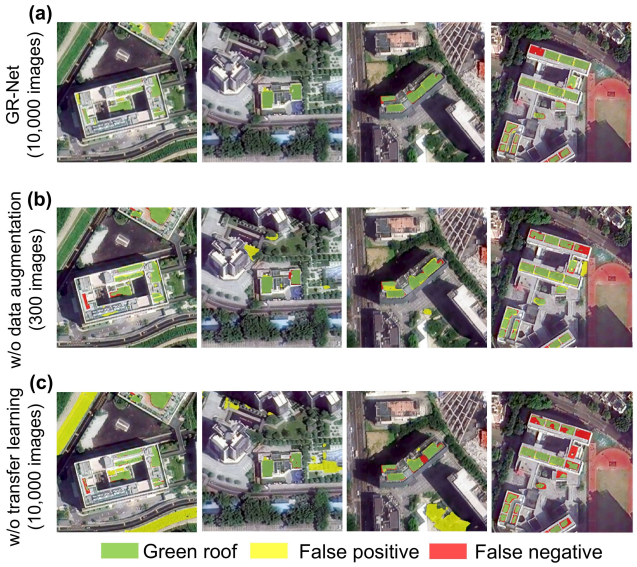


Fig. 9. Comparison of the training dataset sample sizes. (a) Prediction results of GR-Net. (b) and (c) Results obtained without data augmentation and transfer learning, respectively. The "w/o" prefix means "without."

2) Sample Size of the Training Dataset: To assess the impact of training dataset size on model performance, we train the model using only 300 original samples without data augmentation, and its performance is 6%–12% lower than GR-Net. As illustrated in Fig. 9(b), the model struggles to fully

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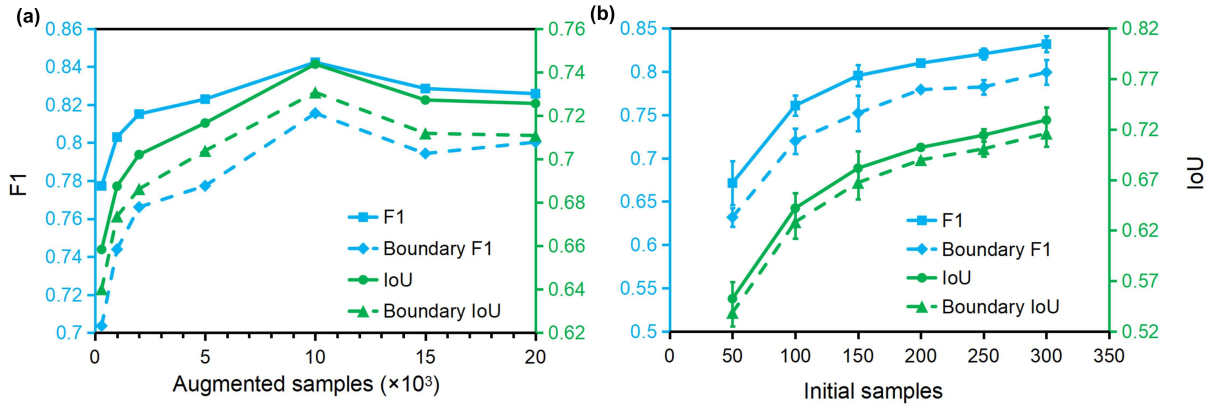


Fig. 10. Effect of training dataset sample size on model performance. (a) and (b) Effectiveness of data augmentation and initial sample size, respectively.

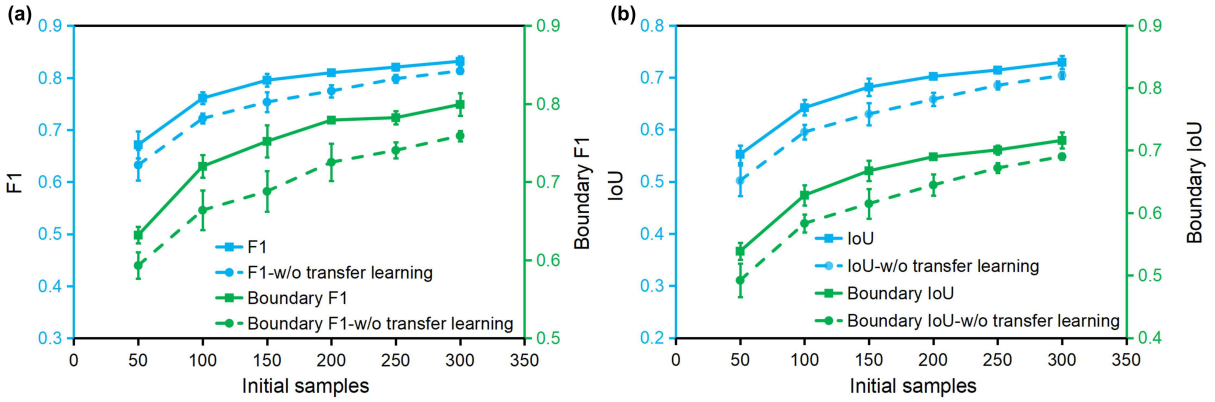


Fig. 11. Effect of transfer learning on model performance. (a) F1 score. (b) IoU. The "w/o" prefix denotes "without."

capture the general features of green roofs when trained on a limited dataset.

To determine the optimal number of augmented training samples, we progressively expand the initial 300 samples to 1000, 2000, 5000, 10 000, 15 000, and 20 000. As shown in Fig. 10(a), model accuracy improves as the number of augmented samples increases, reaching its peak at 10 000 before leveling off and slightly declining. Based on this observation, we set the training sample size to 10 000 in all experiments.

Furthermore, we randomly select 50, 100, 150, 200, 250, and 300 samples from the original 300 training samples and expand each subset to 10 000 using data augmentation. To mitigate randomness, each selection is repeated three times, and the average result is reported. As illustrated in Fig. 10(b), model accuracy steadily improves and stabilizes as the number of initial samples increases.

3) Transfer Learning: To assess the impact of transfer learning on model performance, we train the model from scratch without loading pretrained weights, and its performance is 4%–9% lower than GR-Net. As shown in Fig. 9(c), the absence of transfer learning limits the model's initial feature extraction capability, leading to reduced segmentation accuracy.

To further analyze this effect, we compare model performance with and without pretrained weights across initial training sample sizes of 50, 100, 150, 200, 250, and 300. To mitigate randomness, each selection is repeated three times, and the average prediction is reported as the final result. As illustrated in Fig. 11, the model trained with pretrained weights consistently outperforms its counterpart in all metrics at all sample sizes. This highlights the significant role of transfer learning in enhancing generalization performance.

4) Spatial Resolution of Input Images: We analyze the effect of image resolution on prediction performance by evaluating resolutions of 0.5, 0.75, 1, 1.5, 2, 3, 4, and 5 m. The corresponding image dimensions are 300 × 300, 200 × 200, 150 × 150, 100 × 100, 75 × 75, 50 × 50, 37 × 37, and 30 × 30 pixels, respectively. The quantitative and qualitative results, presented in Figs. 12 and 13, show a significant decline in model performance as resolution decreases. Notably, at a resolution of 5 m, the $F1$ score, IoU, Boundary $F1$ score, and Boundary IoU on the in-domain test dataset decrease by 53.7%, 64.0%, 80.8%, and 67.7%, respectively. This performance deterioration is primarily attributed to the loss of local details at lower resolutions, which hinders the accurate recognition of green roof features and reduces segmentation accuracy.

G. Impact of Thresholds

We evaluate the model performance across a range of threshold values. As shown in Fig. 14, the model achieves

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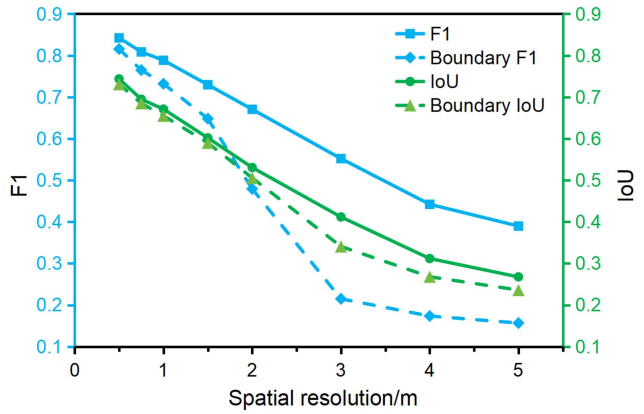


Fig. 12. Effect of spatial resolution on model performance.

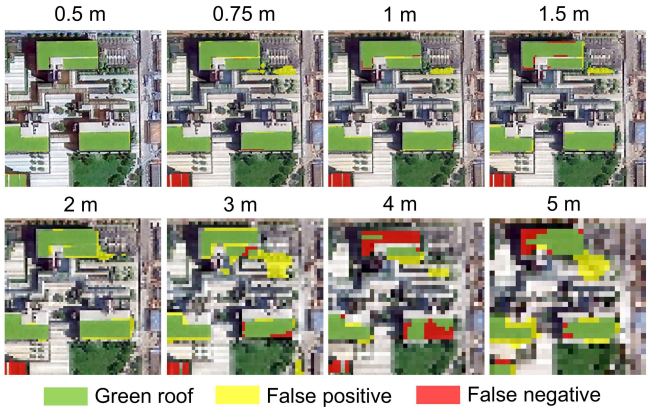


Fig. 13. Comparison of images with different spatial resolutions.

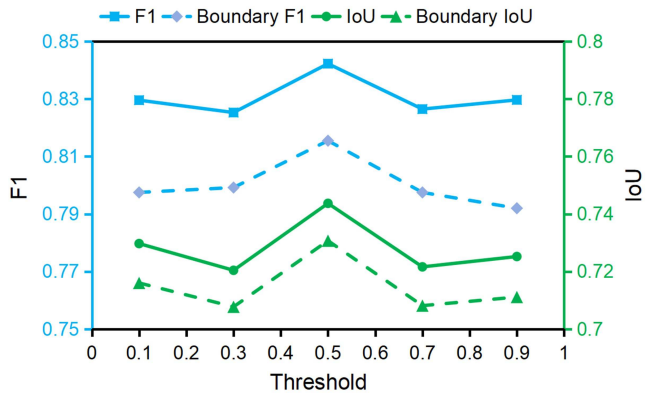


Fig. 14. Effect of threshold on model performance.

its peak performance at a threshold of 0.5. Accordingly, we adapt 0.5 as the fixed threshold in all experiments.

H. Computational Complexity Analysis

We add a comprehensive computational complexity analysis to evaluate the practicality of GR-Net in real-world applications. Specifically, GR-Net contains 30.1 M Params, requires 19.7 G FLOPs, and achieves an inference speed of 28.3 FPS (=35.3 ms/image) on a 300×300 input using a single NVIDIA GTX 4090 GPU.

TABLE IX
GR-NET INFERENCE TIME FOR GREEN ROOFS IN FIVE CITIES

City	Area (km^2)	Inference time (s)
Nanshan	163.7	135.7
Futian	67.3	45.7
Luohu	44.4	38.9
Xicheng	50.6	28.7
Jingan	36.7	22.0

In practical settings, GR-Net can segment green roofs across large-scale urban regions within minutes using parallel processing on a single NVIDIA GTX 4090 GPU, shown in Table IX of the revised manuscript, demonstrating its suitability for real-time and scalable deployment scenarios.

I. Feature Interpretability Analysis

To better illustrate the contribution of each module to green roof segmentation, we use heat maps to visualize the model's intermediate processing steps. As shown in Fig. 15, the process begins by feeding *R*, *G*, and *B* channels into the backbone, which extracts the overall features of the building roofs and green roofs (Feat ①). Meanwhile, remote sensing prior knowledge (NIR and NDVI) is processed through the CBR module, generating feature maps that emphasize green roofs (Feat ②). Next, the model fuses features from both sources using CBAM to obtain Feat ③ that focuses more on green roofs. Then the green roof segmentation head (CCS) outputs a probability distribution map of green roofs (Feat ④). Additionally, the feature map generated by the backbone (Feat ①) is also passed to the building roof segmentation head (CCS) to produce a building roof segmentation mask. To further refine green roof segmentation, this mask is applied to Feat ④, which is filtered to obtain the final segmentation result.

J. Limitations and Future Research

Although GR-Net significantly enhances green roof segmentation accuracy compared to previous methods, it still has certain limitations. First, due to the angle of satellite photography, some green roofs may be partially occluded by building shadows [68], which is not accounted for during the annotation process. Additionally, since the training dataset does not include scenes with water bodies, the model may misegment such areas when predicting green roof distribution at an urban scale. Moreover, the number of supervised training samples is relatively limited due to the high cost of manual annotation.

To further improve GR-Net's applicability across different cities, several enhancements can be explored. First, techniques such as multimodal fusion [69] and image representation learning [70] can address occlusion and mis-segmentation issues by integrating diverse data sources. Second, leveraging latent features and graph neural networks can help model complex relationships and incorporate additional contextual information [71], enabling collaborative filtering between green roofs and building roofs. Finally, adopting semisupervised learning [72], unsupervised learning [73], or adversarial detection [74]

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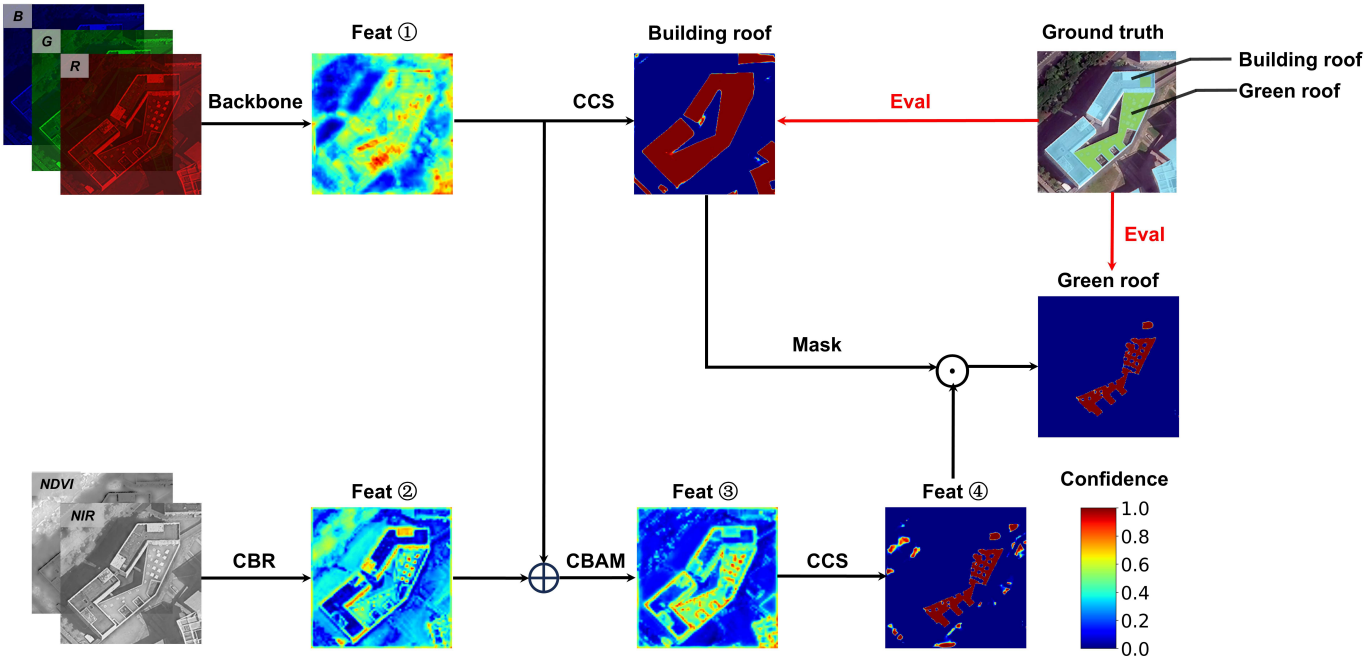


Fig. 15. Heatmap visualization of intermediate processes for semantic segmentation of building roofs and green roofs. The heatmap demonstrates the role of backbone, RSPM, CBAM, and CCS in progressively improving the accuracy of green roof feature representation.

approaches could mitigate the limitations posed by the scarcity of manually annotated data, improving model performance.

V. Conclusion

In this article, we propose a novel end-to-end semantic segmentation framework for green roofs. Using highly urbanized areas of Shenzhen, Beijing, and Shanghai as case studies, we construct a multisource remote sensing dataset containing both green roofs and building roofs. We then introduce GR-Net, a high-resolution attention-based convolutional neural network designed to address the challenges posed by the irregular shapes and sparse distribution of green roofs. To fully leverage the building and vegetation attributes of green roofs, we incorporate the BGM and RSPM modules, which effectively capture vegetation features and filter out ground vegetation in nonroof areas. Additionally, we employ data augmentation and transfer learning to mitigate the limitations imposed by the scarcity of supervised samples. Comprehensive experiments demonstrate that our proposed model achieves outstanding performance on both in-domain and out-of-domain test datasets, highlighting its strong generalization ability across different cities.

In future work, we aim to enhance GR-Net by integrating more advanced techniques, such as multimodal fusion, latent feature modeling, and semisupervised learning to further improve segmentation accuracy. Overall, this work contributes to a more comprehensive study of the ecological benefits of urban-scale green roofs before and after retrofitting, including the effects on thermal environment, carbon sink capacity, and urban runoff.

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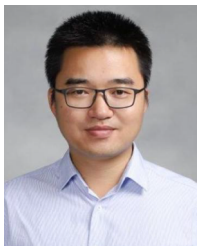
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